

- abstract planning.

Suggested Literature Review Structure

1. Introduction to LLM agents
2. Embodied AI and game environments
3. Natural language grounding
4. Planning and reasoning architectures
5. Memory systems
6. Multi-agent coordination
7. Human-AI interaction in games
8. Current limitations and gaps
9. Proposed research direction

Useful Resources

- [Voyager Project](#)
- [Voyager Paper \(arXiv\)](#)
- [LLM Game Agent Survey](#)
- [Awesome LLM Game Agent Papers](#)

If you want, I can also provide:

- a formal research problem statement,
- a PhD/MS thesis proposal,
- a taxonomy diagram of LLM game agents,
- or a chronological timeline of major papers in this field.

  ...   Sources

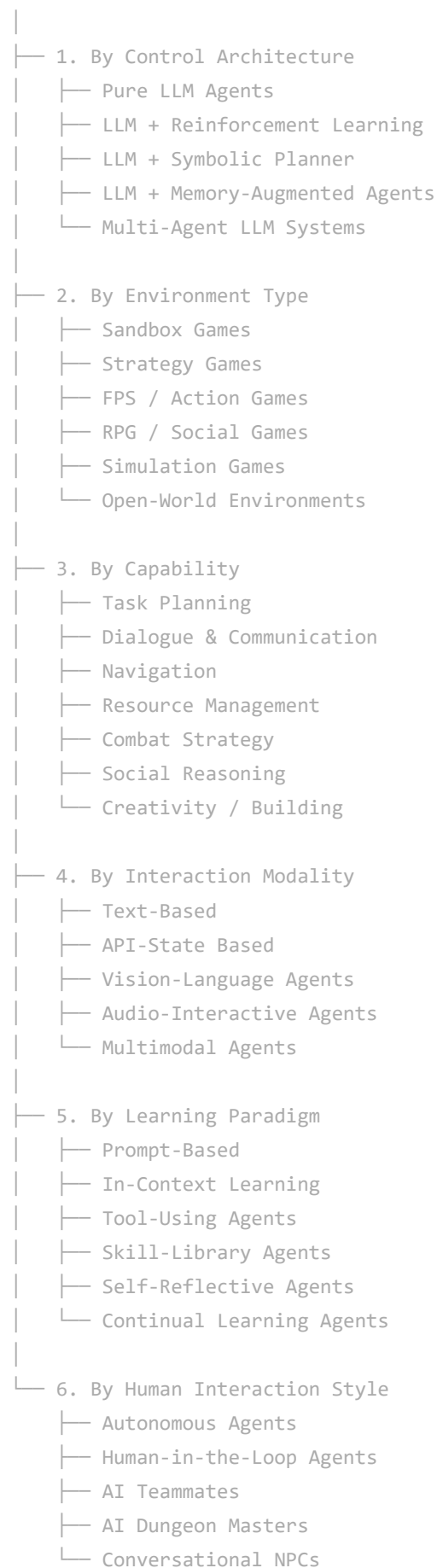
a taxonomy diagram of LLM game agents,

Taxonomy of LLM Game Agents

Below is a structured taxonomy commonly used to organize research on LLM-controlled game agents.

1. High-Level Taxonomy Diagram

LLM Game Agents



2. Detailed Taxonomy Explanation

A. By Control Architecture

1. Pure LLM Agents

The LLM directly:

- interprets commands,
- reasons,
- outputs actions.

Example:

- text adventure agents.

Characteristics

- simple architecture,
- strong reasoning,
- weak low-level control.

2. LLM + Reinforcement Learning Hybrid

Most important modern direction.



Benefits

- strategic reasoning from LLM,
- fast reactions from RL.

Research Example

- hierarchical game AI systems.

3. LLM + Symbolic Planner

Uses:

- classical planning,
- behavior trees,
- symbolic reasoning.

Strength

- reliable long-horizon execution.

Weakness

- less flexible.

4. Memory-Augmented Agents

Adds:

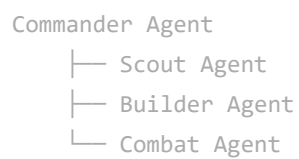
- episodic memory,
- vector databases,
- skill retrieval systems.

Example:

- Voyager.

5. Multi-Agent LLM Systems

Several AI agents collaborate.



Used in:

- RTS,
- MMORPG,
- social simulations.

B. By Environment Type

Sandbox Games

Examples:

- Minecraft

- Terraria

Most popular research testbeds.

Why?

- open-ended tasks,
- creativity,
- resource planning.

Strategy Games

Examples:

- StarCraft
- Civilization

Focus:

- planning,
- coordination,
- economy management.

FPS / Action Games

Examples:

- Doom
- Valorant-like environments

Challenges:

- real-time reaction,
- aiming,
- latency.

RPG / Social Games

Examples:

- Skyrim-like worlds,
- AI villages.

Focus:

- dialogue,
- personality,

- role-play.

Simulation Games

Examples:

- Sims,
- city-building games.

Focus:

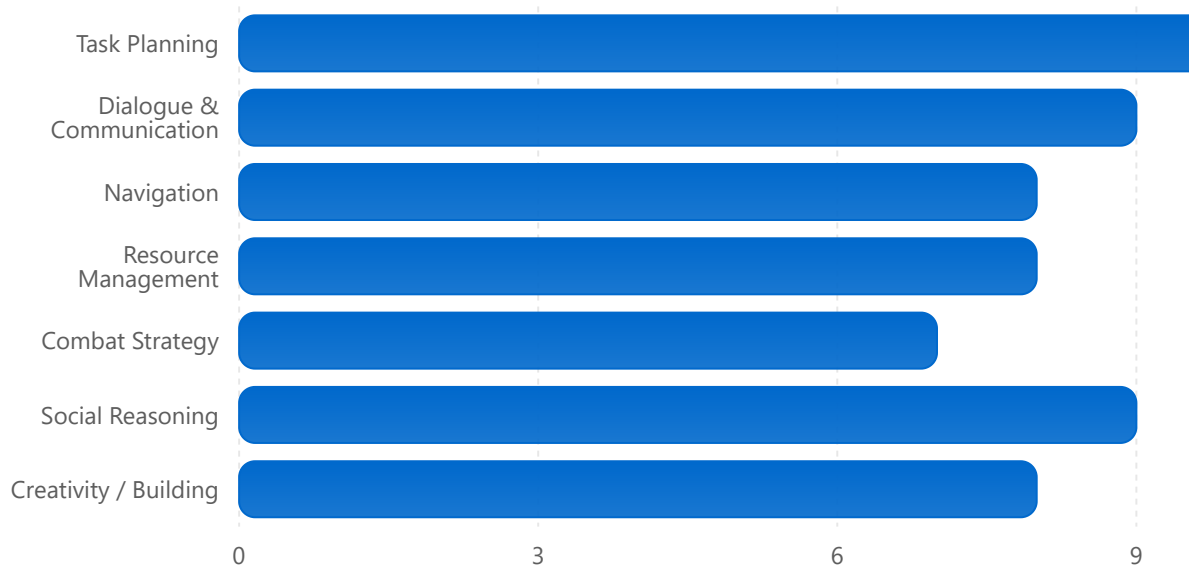
- social interaction,
- economic reasoning.

C. By Capability

Here is a capability-oriented taxonomy.

Core Capability Areas of LLM Game Agents

Common research capability dimensions for LLM-controlled game agents.



Task Planning

The agent:

- decomposes goals,
- sequences actions,

- handles sub-goals.

Example:

“Build a base before nightfall.”

Dialogue & Communication

Used for:

- NPC interaction,
- teammate coordination,
- negotiation.

Social Reasoning

Advanced emerging area.

Includes:

- trust,
- deception,
- alliances,
- emotions.

Creativity / Building

Important in sandbox games.

Example:

- generating structures,
- designing environments.

D. By Interaction Modality

Text-Based Agents

Input/output entirely textual.

Common in:

- text adventures,
- role-playing environments.

API-State Agents

Receive structured game state.

```
{  
  "health": 72,  
  "inventory": ["wood", "stone"],  
  "enemy_distance": 14  
}
```

Most current research uses this.

Vision-Language Agents

Use:

- screenshots,
- visual perception,
- scene understanding.

Major future direction.

Multimodal Agents

Combine:

- vision,
- text,
- audio,
- memory.

Closest to human gameplay.

E. By Learning Paradigm

Prompt-Based Agents

No training.

Behavior entirely from prompts.

Skill-Library Agents

Store reusable skills.

Example:

```
Skill:  
"Craft Iron Pickaxe"
```

Voyager introduced this strongly.

Self-Reflective Agents

Agent critiques itself.

```
Action failed → Analyze failure → Revise strategy
```

Important for autonomous improvement.

Continual Learning Agents

Learn persistently over time.

Still largely unsolved.

F. By Human Interaction Style

Autonomous Agents

Operate independently.

Example:

- self-playing AI.

AI Teammates

Assist human players.

Example:

“Cover me while I reload.”

Very important commercial direction.

Conversational NPCs

Natural dialogue with players.

Used for:

- immersion,
- dynamic storytelling.

3. Emerging Research Directions

Most Active Current Areas

Area	Research Intensity
Memory-Augmented Agents	Very High
Vision-Language Game Agents	Very High
Multi-Agent Cooperation	Very High
Human-like NPCs	High
Real-Time LLM Agents	High
Generalizable Game Agents	High

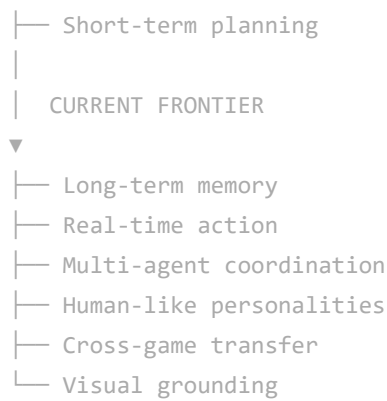
4. Research Gap Map

Solved Better

|

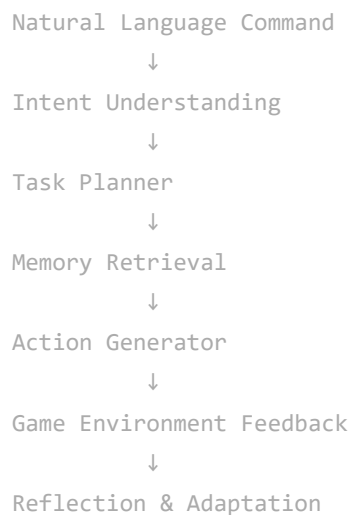
└ Simple task execution

└ Dialogue generation



5. Recommended Conceptual Framework for Your Research

If you are planning a thesis or paper, a strong framework is:



This architecture aligns with:

- Voyager,
- Generative Agents,
- embodied AI systems,
- modern autonomous game AI research.

6. Key Papers by Taxonomy Area

Area	Important Papers
Memory Agents	Voyager
Social Agents	Generative Agents

Area	Important Papers
Multi-Agent Systems	GameGPT
Embodied AI	MineDojo
Vision-Language	DeepMind SIMA
Surveys	LLM Game Agent Survey

Key works:

- [Generative Agents](#)
- [Embodied AI](#)
- [MineDojo](#)
- [DeepMind](#)

If you want next, I can also provide:

1. A visual graphical taxonomy figure (publication-style)
2. A chronological evolution timeline (2020–2026)
3. A comparative table of architectures
4. A detailed research methodology for your thesis
5. Open-source frameworks and datasets for implementation

